National
Qualifications
2026

# 2026 Applications of Mathematics

## National 5 Paper 1

## Question Paper Finalised Marking Instructions

These marking instructions have been prepared by examination teams for use by Qualifications Scotland appointed markers when marking external course assessments.

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**General marking principles for National 5 Applications of Mathematics**

*Always apply these general principles. Use them in conjunction with the detailed marking instructions, which identify the key features required in candidates' responses.*

*For each question, the marking instructions are generally in two sections:*

*generic scheme — this indicates why each mark is awarded*
*illustrative scheme — this covers methods which are commonly seen throughout the marking*

*In general, you should use the illustrative scheme. Only use the generic scheme where a candidate has used a method not covered in the illustrative scheme.*

**(a)** Always use positive marking. This means candidates accumulate marks for the demonstration of relevant skills, knowledge and understanding; marks are not deducted for errors or omissions.

**(b)** If you are uncertain how to assess a specific candidate response because it is not covered by the general marking principles or the detailed marking instructions, you must seek guidance from your team leader.

**(c)** One mark is available for each ○. There are no half marks.

**(d)** If a candidate's response contains an error, all working subsequent to this error must still be marked. Only award marks if the level of difficulty in their working is similar to the level of difficulty in the illustrative scheme.

**(e)** Only award full marks where the solution contains appropriate working. A correct answer with no working receives no mark, unless specifically mentioned in the marking instructions.

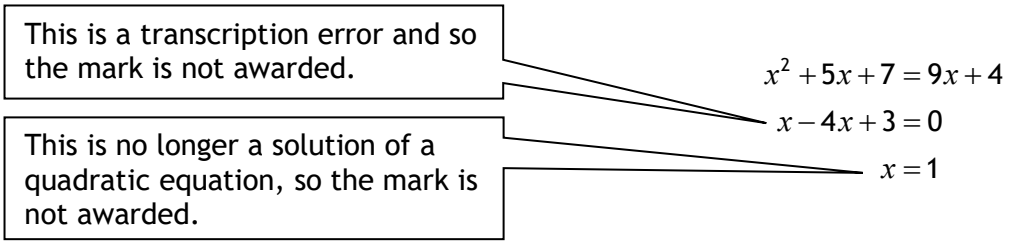
**(f)** Candidates may use any mathematically correct method to answer questions, except in cases where a particular method is specified or excluded.

**(g)** If an error is trivial, casual or insignificant, for example 6 × 6 = 12, candidates lose the opportunity to gain a mark, except for instances such as the second example in point (h) below.

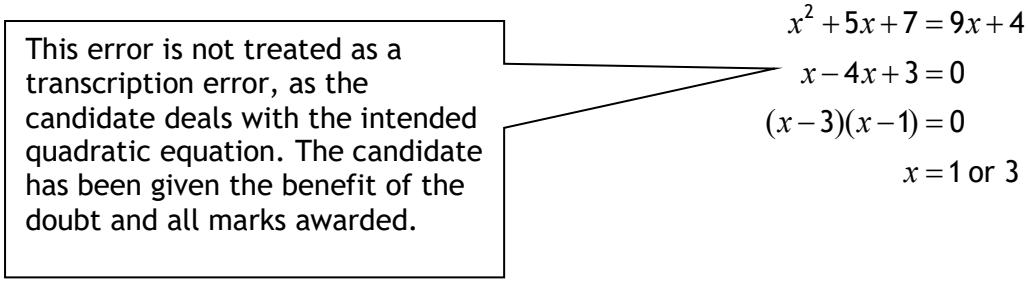


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**(h)** If a candidate makes a transcription error (question paper to script or within script), they lose the opportunity to gain the next process mark, for example




The following example is an exception to the above




**(i) Horizontal/vertical marking**

If a question results in two pairs of solutions, apply the following technique, but only if indicated in the detailed marking instructions for the question.

Example:

| | $\bullet^5$ | $\bullet^6$ |
|---|---|---|
| $\bullet^5$ | $x = 2$ | $x = -4$ |
| $\bullet^6$ | $y = 5$ | $y = -7$ |

Horizontal: $\bullet^5$ $x = 2$ and $x = -4$ Vertical: $\bullet^5$ $x = 2$ and $y = 5$
$\bullet^6$ $y = 5$ and $y = -7$ $\bullet^6$ $x = -4$ and $y = -7$

You must choose whichever method benefits the candidate, **not** a combination of both.

**(j)** In final answers, candidates should simplify numerical values as far as possible unless specifically mentioned in the detailed marking instruction. For example

$\frac{15}{12}$ must be simplified to $\frac{5}{4}$ or $1\frac{1}{4}$ $\frac{43}{1}$ must be simplified to 43

$\frac{15}{0{\cdot}3}$ must be simplified to 50 $\frac{4/5}{3}$ must be simplified to $\frac{4}{15}$

$\sqrt{64}$ must be simplified to 8*

*The square root of perfect squares up to and including 144 must be known.



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**(k)** Commonly Observed Responses (COR) are shown in the marking instructions to help mark common and/or non-routine solutions. CORs may also be used as a guide when marking similar non-routine candidate responses.

**(l)** Do not penalise candidates for any of the following, unless specifically mentioned in the detailed marking instructions:

- working subsequent to a correct answer
- correct working in the wrong part of a question
- legitimate variations in numerical answers/algebraic expressions, for example angles in degrees rounded to nearest degree
- omission of units
- bad form (bad form only becomes bad form if subsequent working is correct), for example

  $(x^3+2x^2+3x+2)(2x+1)$ written as

  $(x^3+2x^2+3x+2)\times 2x+1$

  $=2x^4+5x^3+8x^2+7x+2$

  gains full credit

- repeated error within a question, but not between questions or papers

**(m)** In any 'Show that…' question, where candidates have to arrive at a required result, the last mark is not awarded as a follow-through from a previous error, unless specified in the detailed marking instructions.

**(n)** You must check all working carefully, even where a fundamental misunderstanding is apparent early in a candidate's response. You may still be able to award marks later in the question so you must refer continually to the marking instructions. The appearance of the correct answer does not necessarily indicate that you can award all the available marks to a candidate.

**(o)** You should mark legible scored-out working that has not been replaced. However, if the scored-out working has been replaced, you must only mark the replacement working.

**(p)** If candidates make multiple attempts using the same strategy and do not identify their final answer, mark all attempts and award the lowest mark. If candidates try different valid strategies, apply the above rule to attempts within each strategy and then award the highest mark.

For example:

| | |
|---|---|
| Strategy 1 attempt 1 is worth 3 marks. | Strategy 2 attempt 1 is worth 1 mark. |
| Strategy 1 attempt 2 is worth 4 marks. | Strategy 2 attempt 2 is worth 5 marks. |
| From the attempts using strategy 1, the resultant mark would be 3. | From the attempts using strategy 2, the resultant mark would be 1. |

In this case, award 3 marks.



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**Marking Instructions for each question**

| Question | | | Generic scheme | Illustrative scheme | Max mark |
|---|---|---|---|---|---|
| **1.** | | | •[1] Strategy/communication: mark one pressure correctly on scale<br><br>•[2] Communication: mark other pressure on scale and conclusion consistent with working | •[1] 8.2 or 112<br><br>•[2] 112 or 8.2 and not safe | **2** |

**Notes:**

1. Where a candidate has marked no pressures on the diagram. award 0/2
2. •[2] is not available where a candidate has marked more than two pressures.
3. Full marks can only be awarded if there are two pressures marked on the diagram.
4. Do not penalise incorrect attempts at a numerical conversion subsequent to two correct pressures marked on scale and a correct conclusion. See COR1.

**Commonly Observed Responses:**

1. 8.2 and 112 both marked correctly→Not safe because 112=7.4 award 2/2 ✓✓



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| Question | | | Generic scheme | Illustrative scheme | Max mark |
|---|---|---|---|---|---|
| **2.** | | | $\bullet^1$ Process: evidence of common denominator<br><br>$\bullet^2$ Process: calculate numerators and add fractions<br><br>$\bullet^3$ Process: calculate fraction of sheep who have not yet had their dose of vitamin | $\bullet^1$ $\frac{\ldots}{21}$ or equivalent<br><br>$\bullet^2$ $\left(\frac{3}{21}+\frac{14}{21}=\right)\frac{17}{21}$<br><br>$\bullet^3$ $\frac{4}{21}$ | **3** |

**Notes:**

1. Correct answer without working. award 0/3
2. The final answer does not need to be in its simplest form.
3. Do not penalise incorrect simplification of final answer.
4. $\bullet^3$ is not available when an improper fraction is calculated for $\bullet^2$, with the exception of COR1.

**Commonly Observed Responses:**

1. $\frac{\ldots}{21} \rightarrow \left(\frac{6}{7}+\frac{1}{3}=\right)\frac{25}{21} \rightarrow \frac{4}{21}$ award 3/3 ✓✓✓
2. $\frac{\ldots}{21} \rightarrow \frac{1}{21}+\frac{2}{21}=\frac{3}{21} \rightarrow \frac{18}{21}$ or $\frac{6}{7}$ award 2/3 ✓×✓
3. $\left(\frac{1}{7}\times\frac{2}{3}=\right)\frac{\ldots}{21} \rightarrow \frac{2}{21} \rightarrow \frac{19}{21}$ award 2/3 ✓×✓
4. $\frac{\ldots}{10} \rightarrow \frac{3}{10}+\frac{14}{10}=\frac{17}{10} \rightarrow$ any final answer award 1/3 ×✓×
5. $\frac{\ldots}{10} \rightarrow \frac{2}{10}+\frac{1}{10}=\frac{3}{10} \rightarrow \frac{7}{10}$ award 1/3 ××✓



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| Question | | | Generic scheme | Illustrative scheme | Max Mark |
|---|---|---|---|---|---|
| **3.** | | | $\bullet^1$ Process: calculate how much Helen will save each month<br><br>$\bullet^2$ Process/communication: calculate the minimum number of months and round appropriately | $\bullet^1$ $(20\% \text{ of } 350 =) 70$<br><br>$\bullet^2$ $(1250 \div 70 = 17.8... =) 18$ (months) | **2** |

**Notes:**

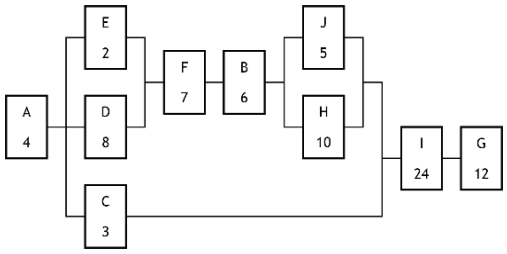
1. Correct answer without working award 0/2
2. $\bullet^2$ is still available where the working for the division is incomplete eg $1250 \div 70 = 17 \cdot ...$ (or just 17) $\rightarrow 18$.
3. $\bullet^2$ can be awarded where the working shows $18 \times 70 = 1260$ followed by a final answer of 18.
4. $\bullet^2$ is not available where the candidate's answer to $\bullet^1$ is a factor of 1250.
5. Ignore errors in calculations after the decimal point or in remainder.
6. Where a candidate uses a strategy involving repeated addition or subtraction of 70, $\bullet^2$ can only be awarded where all calculations are correct.
7. Ignore omitted or incorrect units.

**Commonly Observed Responses:**

1. $70 \rightarrow 1250 \div 70 = 18$ award 2/2 ✓✓
2. $350 \rightarrow (1250 \div 350 =) 4$ award 1/2 ✗✓



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| Question | | | Generic scheme | Illustrative scheme | Max mark |
|---|---|---|---|---|---|
| **4.** | (a) | | •[1] Strategy/communication: any 5 task letters and times in the correct boxes<br>•[2] Strategy/communication: complete diagram | •[1&2]   | **2** |
| **Notes:**<br>1. C is **not** interchangeable with D or E.<br>2. D and E are interchangeable.<br>3. H and J are interchangeable.<br>4. If a candidate omits all times, all task letters must be correct to be awarded 1/2.<br>5. If a candidate omits all task letters, all times must be correct to be awarded 1/2. | | | | | |
| **Commonly Observed Responses:** | | | | | |
| | (b) | | •[3] Process/communication: state minimum time | •[3] 71 (hours) | **1** |
| **Notes:**<br>1. Correct answer without working. award 1/1<br>2. Ignore omitted or incorrect units. | | | | | |
| **Commonly Observed Responses:** | | | | | |



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| Question | | | Generic scheme | Illustrative scheme | Max mark |
|---|---|---|---|---|---|
| **5.** | (a) | | $\bullet^1$ Process: determine the median<br><br>$\bullet^2$ Process: determine the upper and lower quartiles | $\bullet^1$ 17.5<br><br>$\bullet^2$ 10, 26 | **2** |

**Notes:**

Ordered list: 7, 9, 10, 14, 17, 18, 21, 26, 27, 31

1. The answers for (a) can be inferred from the boxplot.
2. If obtaining median and quartiles from the original unordered list, only $\bullet^2$ is available, see COR3.
3. If one number is missed from or added to an **unordered** list. award 0/2
4. If one number is missed from or added to an **ordered** list:
   - $\bullet^1$ is not available,
   - $\bullet^2$ is only available when all three quartiles are consistent with the candidate's list.
5. If **more than one** number is missed from or added to any list. award 0/2
6. Do not penalise incorrect labelling of 17.5, 10 or 26, eg a candidate who says LQ = 26 or UQ = 10.

**Commonly Observed Responses:**

1. 7 9 10 14 17 18 21 26 27 31 with no labels award 2/2 ✓✓
   10 17.5 26
2. 7, 10, 17.5, 26, 31 award 2/2 ✓✓
3. 24.5, 27, 7 (from an unordered list) award 1/2 ×✓

| Question | | | Generic scheme | Illustrative scheme | Max mark |
|---|---|---|---|---|---|
| 5. | (b) | | •[3] Communication: mark end points correctly<br><br>•[4] Communication: mark $Q_1$, $Q_2$, $Q_3$ and draw box plot | •[3] end points at 7 and 31<br><br>•[4] box showing $Q_1$, $Q_2$, $Q_3$ | **2** |

**Notes:**

1. The answers for (a) can be inferred from the boxplot.
2. If the candidate constructs a dot plot. award 0/2
3. When a point falls between 2 divisions, accept a mark at any value in between the divisions.
4. •[3&4] are still available when the calculated value of $Q_2$ does not lie between the calculated values of $Q_1$ and $Q_3$.
5. •[4] is not available if:
   a. more than 5 values are marked.
   b. the five-figure summary is marked using only dots or crosses.

**Commonly Observed Responses:**

1. In Part (a): 17.5, 7, 31
   Followed by in part (b): correct boxplot drawn using 7, 10, 17.5, 26, 31
   For part (a), award 1/2 ✓×
   For part (b), award 2/2 ✓✓

| | | | | | |
|---|---|---|---|---|---|
| | (c) | | •[5] Process: calculate interquartile range | •[5] $(26-10=)16$ | **1** |

**Notes:**

1. •[5] must be consistent with previous working.

**Commonly Observed Responses:**



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| Question | | | Generic scheme | Illustrative scheme | Max mark |
|---|---|---|---|---|---|
| **6.** | | | $\bullet^1$ Process: convert to consistent units<br><br>$\bullet^2$ Process/communication: express gradient as a fraction, **and** write in its simplest form | $\bullet^1$ 1.8 (m) **or** 1200 (cm)<br><br>$\bullet^2$ $\left(\frac{180}{1200}=\right)\frac{3}{20}$ | **2** |

**Notes:**

1. Correct answer without working. award 2/2
2. $\bullet^2$ is not available when an attempt is made to further simplify a fraction already in its simplest form eg $\frac{3}{20}=\frac{1}{4}$.
3. $\bullet^2$ is not available where a candidate only gives their gradient as a decimal or a percentage.

**Commonly Observed Responses:**

1. 180 and 120 $\rightarrow\left(\frac{180}{120}=\right)\frac{3}{2}$ award 1/2 ×✓
2. 180 and 120 $\rightarrow\left(\frac{120}{180}=\right)\frac{2}{3}$ award 0/2 ××
3. 180 and 12 $\rightarrow\left(\frac{180}{12}=\right)15$ award 1/2 ×✓
4. 180 and 12 $\rightarrow\left(\frac{180}{12}=\right)\frac{15}{1}$ award 0/2 ××
5. 180 and 12 $\rightarrow\left(\frac{12}{180}=\right)\frac{1}{15}$ award 0/2 ××



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| Question | | | Generic scheme | Illustrative scheme | Max mark |
|---|---|---|---|---|---|
| **7.** | | | $\bullet^1$ Process: calculate commission<br><br>$\bullet^2$ Process: calculate gross pay | $\bullet^1$ $(5\%$ of $16000 =)800$<br><br>$\bullet^2$ (1765 + 800 =) (£)2565 | **2** |

**Notes:**

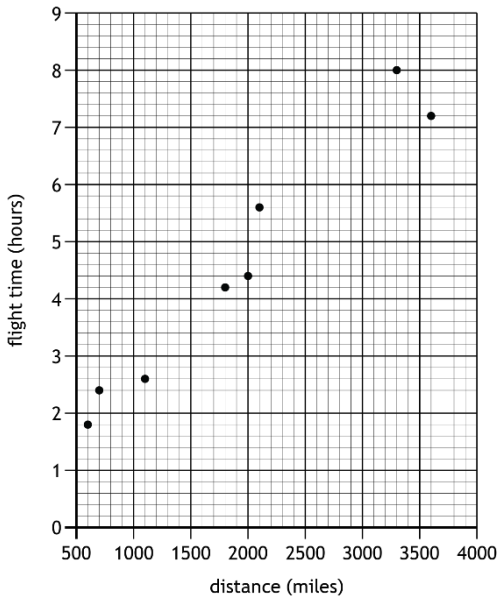
1. Correct answer without working. award 2/2
2. $\bullet^2$ is only available for adding 1765 to a previously calculated commission.
3. Where the final answer is not a whole number, $\bullet^2$ is only available where the final answer is rounded or truncated to 2 decimal places.

**Commonly Observed Responses:**

1. $(5\%$ of $56000 =)2800 \rightarrow 4565$ award 1/2 ×✓
2. $(5\%$ of $40000 =)2000 \rightarrow 3765$ award 1/2 ×✓
3. $(5\%$ of $1765 =)88.25 \rightarrow 1853.25$ award 1/2 ×✓



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| Question | | | Generic scheme | Illustrative scheme | Max mark |
|---|---|---|---|---|---|
| **8.** | (a) | | •[1] Communication: any 4 points plotted correctly<br><br>•[2] Communication: remaining 4 points plotted correctly | •[1&2]   | **2** |

**Notes:**

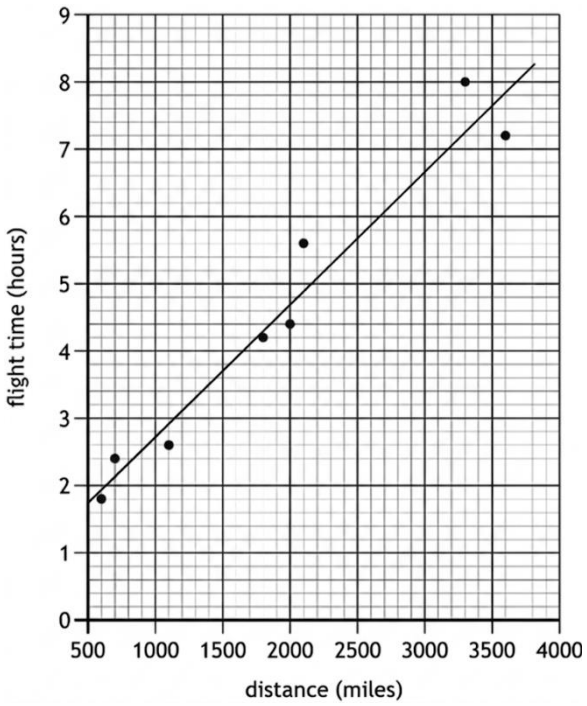
| **Distance (miles)** | 600 | 700 | 1100 | 1800 | 2000 | 2100 | 3300 | 3600 |
|---|---|---|---|---|---|---|---|---|
| **Flight time (hours)** | 1.8 | 2.4 | 2.6 | 4.2 | 4.4 | 5.6 | 8.0 | 7.2 |

1. •[1] and •[2] are still available if a candidate constructs a line graph.
2. •[2] is not available where a candidate plots an extra point (with the exception of any point connected to 3.4 hours and part (c)).
3. If a candidate constructs a bar graph. award 0/2

**Commonly Observed Responses:**



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| Question | | | Generic scheme | Illustrative scheme | Max mark |
|---|---|---|---|---|---|
| **8.** | (b) | | $\bullet^3$ Strategy: draw consistent line of best fit | $\bullet^3$   | **1** |

**Notes:**

1. The line of best fit must be straight and have a positive gradient.
2. $\bullet^3$ is not available if the line of best fit goes through the origin or the x-axis.
3. $\bullet^3$ is not available if there are no points below the line of best fit.
4. $\bullet^3$ is not available if there are no points above the line of best fit.
5. $\bullet^3$ is not available if a line graph or bar graph are plotted.
6. $\bullet^3$ is not available if the line of best fit does not extend across the full range of plotted distances eg 600 and 3600.

**Commonly Observed Responses:**

| | | | | | |
|---|---|---|---|---|---|
| | (c) | | $\bullet^4$ Strategy/communication: state estimated distance for 3.4 hours consistent with line of best fit | $\bullet^4$ eg 1400 (miles) | **1** |

**Notes:**

1. When the distance falls between 2 divisions, accept either number or any value in between.
2. $\bullet^4$ is still available for a candidate who draws a line graph for $\bullet^3$.
3. $\bullet^4$ is not available where no attempt has been made to draw a line of best fit.

**Commonly Observed Responses:**



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| Question | | | Generic scheme | Illustrative scheme | Max mark |
|---|---|---|---|---|---|
| **9.** | | | •[1] Process: convert time to seconds<br><br>•[2] Process/communication: calculate flow rate as a decimal | •[1] 90<br><br>•[2] $\left(54 \div 90 =\right) 0.6$ ($m^3/s$) | **2** |

**Notes:**

1. Correct answer without working. award 2/2
2. •[2] is not available to a candidate who leaves their answer as a fraction.

**Commonly Observed Responses:**

1. $1.5 \rightarrow \left(54 \div 1.5 =\right) 36$ leading to $\left(36 \div 60 =\right) 0.6$ award 2/2 ✓✓
2. $1.5 \rightarrow (54 \div 1.5 =) 36$ award 1/2 ×✓
3. $90 \rightarrow (54 \div 90 =) 0.6^3$ award 1/2 ✓×



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| Question | | | Generic scheme | Illustrative scheme | Max mark |
|---|---|---|---|---|---|
| **10.** | | | •[1] Strategy/process: calculate cost for a suitable scaling<br><br>•[2] Process: calculate the cost of 1 kg | •[1] eg 1 g = (£)0.0096<br>50 g = (£)0.48<br>100 g = (£)0.96<br><br>•[2] (£)9.60 | **2** |

**Notes:**

1. Correct answer without working. award 0/2
2. •[1] can only be awarded for calculating the cost for **any** weight that is a factor or multiple of 1000 g.
3. Where •[1] is not awarded, •[2] is only available if it is clear from working which weight is linked with the cost in •[1].
4. •[2] may be awarded for adding an incorrect **calculated** cost for 100 grams to (£)8.64, see COR1.
5. Where the final answer is not a whole number, •[2] is only available where the final answer is rounded or truncated to 2 decimal places.

**Commonly Observed Responses:**

1. $100\text{ g} = 0.43 \rightarrow (8.64 + 0.43 = £)9.07$ award 1/2 ×✓
2. 900 g = (£)8.64 → no further valid working award 0/2 ××
3. 4.32 or 43.2(0) with or without working award 0/2 ××



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| Question | Generic scheme | Illustrative scheme | Max mark |
|---|---|---|---|
| **11.** | **Method 1** | | **3** |
| | $\bullet^1$ Strategy/communication: identify and label all the possible combinations | $\bullet^1$ eg<br>1 2 3 4 5 6<br>T (or 1)<br>H (or 2) | |
| | $\bullet^2$ Process: identify all winning combinations | $\bullet^2$ eg<br>1 2 3 4 5 6<br>T (or 1) ✓ ✓<br>H (or 2) ✓ ✓ ✓ | |
| | $\bullet^3$ Communication: state probability of winning | $\bullet^3$ $\frac{5}{12}$ | |
| | **Method 2** | | |
| | $\bullet^1$ Strategy/communication: identify and label fraction of winning combinations for Tails and Heads | $\bullet^1$ $\frac{2}{6}$ labelled with '1.', 'T' or 'Tails' **and** $\frac{3}{6}$ labelled with '2.', 'H' or 'Heads' | |
| | $\bullet^2$ Strategy/communication: identify the four fractions that need to be combined | $\bullet^2$ $\frac{2}{6}$ with $\frac{1}{2}$ **and** $\frac{3}{6}$ with $\frac{1}{2}$ | |
| | $\bullet^3$ Process: state probability of winning | $\bullet^3$ $\left(\frac{2}{6}\times\frac{1}{2}+\frac{3}{6}\times\frac{1}{2}=\right)\frac{5}{12}$ | |

**Notes:**

1. $\frac{5}{12}$ with no other working. award 0/3 ××××
2. Where a 6×6 table is used. award 0/3 ××××
3. The final answer does not need to be in its simplest form.
4. $\bullet^1$ is only available with appropriate labelling ('T'/'H', 'Tails'/'Heads', '1.'/'2.'). Where labels are omitted, $\bullet^2$ is still available. See CORs 4 and 5.
5. For $\bullet^3$:
   a) Accept: 5 out of 12 5 in 12
   b) Do not accept: 5:12 5 to 12 5-12
6. $\bullet^3$ is not available where a candidate gives their answer as **both** a fraction and a ratio, see COR1.
7. In method 1, $\bullet^2$ can be awarded for listing the 5 winning combinations (eg Tails 3&6, Heads 1,3&5). See CORs 2&3.
8. In method 1, $\bullet^1$ may be implied by $\bullet^3$. See COR2.



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| Question | Generic scheme | Illustrative scheme | Max mark |
|---|---|---|---|

**11. (continued)**

**Commonly Observed Responses:**

1. $\frac{5}{12}$ and $5:12$ both stated (following appropriate working) — award 2/3 ✓✓×

2. Tails 3, 6
   Heads 1, 3, 5
   leading to $\frac{5}{12}$
   award 3/3 ✓✓✓

3. Tails 3, 6
   Heads 1, 3, 5
   with no further valid working
   award 1/3 ×✓×

4. Insufficient labelling for $\bullet^1$ leading to correct answer (method 1)

| 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|
| | | | | | |
| | | | | | |

$\rightarrow$

| 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|
| | | ✓ | | | ✓ |
| ✓ | | ✓ | | ✓ | |

$\rightarrow \frac{5}{12}$

award 2/3 ×✓✓

5. Insufficient labelling for $\bullet^1$ leading to correct answer (method 2)

   $\frac{2}{6}$ and $\frac{3}{6}$ with no labelling $\rightarrow$ $\frac{2}{6}$ with $\frac{1}{2}$ and $\frac{3}{6}$ with $\frac{1}{2}$ $\rightarrow \frac{5}{12}$

   award 2/3 ×✓✓

6. Tails: $\frac{2}{6}$
   Heads: $\frac{3}{6}$
   $\rightarrow$ $\frac{2}{6}+\frac{1}{2}=\ldots$ and $\frac{3}{6}+\frac{1}{2}=\ldots$ $\rightarrow$ any answer
   award 2/3 ✓✓×



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| Question | | | Generic scheme | Illustrative scheme | Max mark |
|---|---|---|---|---|---|
| **12.** | | | •1 Process: calculate the arc length of the quarter circle<br><br>•2 Process: calculate the length of the fence | •1 15.7<br><br>•2 $(15.7+10+10+10+9=)54.7$ (m) | **2** |

**Notes:**

1. Correct answer without working. award 0/2
2. Where candidate uses $\pi r^2$. award 0/2
3. •2 is only available for adding 39 to a previously calculated arc length.

**Commonly Observed Responses:**

1. $3.14\times5=15.7\rightarrow54.7$ award 2/2 ✓✓
2. $(3.14\times20=)62.8\rightarrow101.8$ award 1/2 ×✓
3. $(3.14\times20\div2=)31.4\rightarrow70.4$ award 1/2 ×✓
4. $(3.14\times10\div4=)7.85\rightarrow46.85$ award 1/2 ×✓
5. $(3.14\times5\div4=)3.925\rightarrow42.925$ award 1/2 ×✓



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| Question | | | Generic scheme | Illustrative scheme | Max mark |
|---|---|---|---|---|---|
| **13.** | | | **Method 1: comparing proportions**<br>•1 Strategy/communication: state fraction for 'dislike' in one survey<br>•2 Strategy/process: express both 'dislike' fractions in comparable form<br>•3 Communication: state conclusion consistent with working | •1 $\frac{60}{360}$ **or** $\frac{70}{480}$ **or** equivalent<br>•2 eg $\frac{8}{48}$ **and** $\frac{7}{48}$ **or** 0.16(66…) **and** 0.14(58…)<br>•3 'yes' **or** 'decreased' | **3** |
| | | | **Method 2: comparing number of people**<br>•1 Strategy/communication: state fraction of 'dislike' in first survey<br>•2 Process: calculate fraction of 480<br>•3 Communication: state conclusion consistent with working | •1 $\frac{60}{360}$ **or** equivalent<br>•2 $(480 \div 6 =) 80$<br>•3 'yes' **or** 'decreased', with evidence of comparing with 70 | |
| | | | **Method 3: comparing angles**<br>•1 Strategy/communication: state fraction of 'dislike' in second survey or calculate the angle per person<br>•2 Process: calculate angle for 'dislike' in second survey<br>•3 Communication: state conclusion consistent with working | •1 $\frac{70}{480}$ or $(360 \div 480 =) 0.75(^\circ)$<br>•2 $(0.75 \times 70 =) 52.5(^\circ)$<br>•3 'yes' or 'decreased' with evidence of comparing with $60(^\circ)$ | |

**Notes:**

1. 'Yes' or 'decreased' with no working. award 0/3
2. In all methods, •1 can still be awarded when additional fractions are also stated.
3. •1 can be implied by •2 in all methods.
4. Where candidate selects the incorrect angle from the pie chart, •3 is still available.
5. •3 can be awarded where the candidate states a calculated decrease, eg 10 (in method 2) or 7.5 (in method 3).
6. •3 can only be awarded where the candidate makes a conclusion based on comparing 'like with like'.

**Commonly Observed Responses:**

1. $70 \rightarrow 70$ (people) and $60(^\circ) \rightarrow$ 'Yes' or 'decreased' or '10 fewer' award 0/3 ✗✗✗

**Special case:**

2. $480 \div 70$ **and** $360 \div 60 \rightarrow 6.8(57\ldots)$ **and** $6 \rightarrow$ 'Yes' or 'decreased' award 3/3 ✓✓✓
3. $480 \div 70$ **and** $360 \div 60 \rightarrow 6.8(57\ldots)$ **and** $6 \rightarrow$ No conclusion award 2/3 ✓✓✗

**[END OF MARKING INSTRUCTIONS]**



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